Dual-Regime Act/Abort:
Sensor Perturbation vs. Selection Shift

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Abstract

Temperature scaling and related post-hoc maps are often treated as a cheap fix for miscalibrated probabilities. We measure whether that fix transfers under two different test-time changes, for logistic regression, random forests, and histogram gradient boosting on small tabular datasets (no deep nets, no API models). *Feature mean-shift with frozen labels*—a perturbation of X that does **not** preserve $P(Y \mid X)$—raises ECE: at strength 1.5, uncalibrated $\Delta\text{ECE} = \text{ECE}_{\text{pert}} - \text{ECE}_{\text{iid}}$ is positive in 51/60 seed-level cells (mean 0.113, cell-to-cell sd 0.149). That pool mixes sklearn sets (mean 0.042, 25/30 positive) with synthetics (mean 0.184, 26/30 positive) where accuracy also falls. Averaging seeds first, all 12/12 uncalibrated dataset×model means are positive (a sign count; models that share a dataset are not independent). After i.i.d. temperature, isotonic, or histogram maps, pooled ΔECE remains positive. The same maps can help or hurt relative to doing nothing. By contrast, *selection* on X that keeps the original (X, y) pairs—closer to covariate shift, but a milder change in accuracy—yields smaller ΔECE (quantile-slice cell-mean 0.016 on 6 dataset×model cells from 2 datasets). Unweighted split-conformal coverage falls under feature perturbation. On a contact/grasp *act vs. abort* proxy with redundant sensors, a residual router that *switches channels* under perturbation recovers utility (93.0 vs. deployed-channel gating -183.6, 5 seeds) and beats abort-all (0.0). This is an analysis-plus-system result, not a new calibrator and not a physical robot.

1 Introduction

A classifier is *calibrated* when a predicted probability p is correct with frequency p. Post-hoc maps—Platt-style scaling, histogram binning, isotonic regression, and temperature scaling—are the standard way to improve calibration without retraining [1–3]. A separate literature shows that deep-net uncertainty *collapses* under dataset shift [4–6]. Almost all of that evidence is vision-scale neural nets. Tree ensembles remain strong on tabular data [7], and their i.i.d. calibration properties have been known for decades [2]. Within our API-verified bibliography—mostly deep-net calibration and conformal theory, not a systematic search of applied tabular-drift venues—we did not find this exact measurement (sklearn models, i.i.d. post-hoc maps, test-time feature perturbation vs. selection). That is a scoped gap, not a proof of absence.

We treat that as a measurement question. We do not introduce a new calibrator. A negative or mixed answer is an acceptable outcome.

Primary hypothesis (reframed). I.i.d.-fitted post-hoc maps do not keep ECE from rising under a Gaussian mean-shift of test features with *labels held fixed* (input perturbation / a change in $P(Y \mid X)$ at the observed x). We do **not** claim that genuine covariate shift—selection on X that preserves pairings—has the same effect; that is tested as a control.

Two other hypotheses were piloted and *not* selected as the paper claim: isotonic overfit vs. temperature as a function of n_{cal} (H2; heterogeneous), and oracle-weighted conformal recovery (H3; the 1-D weight estimator failed). Decision log: `logs/decisions.md`.

2 Related work

Surveys of calibration methods and metrics include [8, 9]. ECE via confidence binning [3, 10] is biased and definition-dependent [11–13]; we therefore also report Brier score [14] and NLL [15] (pooled Δ alongside ECE in Results). Temperature scaling [3] and Dirichlet calibration [16] target neural nets. Unsupervised recalibration under covariate shift exists for neural models (Pampari and Ermon [17]). Domain-drift post-hoc calibration of CNNs is studied by Tomani et al. [6]. Those papers already *predict* our headline outcomes: ECE should rise under shift, and i.i.d.-fitted post-hoc maps should not be a complete fix. A surprising result here would have been that logistic regression, random forests, and histogram boosting *stay* calibrated under a mean-shift of used covariates *with labels frozen*. Selection that preserves (X, y) pairs is a different mechanism. We reuse the questions, not the model class.

Conformal prediction has finite-sample coverage under exchangeability [18–20] and requires density-ratio weights under covariate shift [21]. We use unweighted split conformal as a secondary diagnostic, not as a proposed fix. Conformal warning systems for grasping and driving [22] and perception-based safe control [23, 24] treat shift as one blob. They do not route different act/abort policies for sensor disagreement vs. workspace selection.

3 Methods

Models. Scikit-learn logistic regression (with standard scaling), random forests, and `HistGradientBoostingClassifier`, with hyperparameters as in `src/calibshift/models.py`. All expose `predict_proba`.

Calibrators. Fitted on a held-out calibration split of the *source* distribution, then frozen. `none`: identity. `temperature`: one scalar T on $\log p$, fitted by NLL (Guo-style, using log-probabilities so trees without logits still apply). `isotonic`: sklearn isotonic regression on $p(y{=}1)$ (binary) or on confidence (multiclass). `histogram`: equal-width binning of the same 1-D score.

Shifts. *Feature perturbation:* add $s{\cdot}\hat{\sigma}_j$ to listed test columns, **keeping the original labels**. This moves x without moving y, so it is *not* covariate shift ($P(Y \mid X)$ is not invariant). $\hat{\sigma}$ is the test-split feature sd. *Selection:* quantile slice and importance resampling of existing (X, y) pairs (closer to covariate shift). Trailing-column perturbation is a control.

Metrics. Equal-width confidence ECE with 15 bins (from the exp04 config), Brier, NLL, accuracy, and split conformal coverage with the inverse-probability (LAC) score $1 - p_y$ at $\alpha = 0.1$ (not APS).

Protocol. Train/calibration/test partitions follow `calibshift.runner` (source-distribution splits; calibration data never includes the test set), stratified, with the 5 seeds listed in the exp04 config. Results are written only by `calibshift.io.write_result`. `paper/numbers.tex` is generated from those JSON files.

4 Experiments

Datasets. Offline-only: sklearn `breast_cancer` (binary) and `wine` (3-class); two `make_classification` sets (binary and 3-class). No OpenML, no paid APIs.

Main grid (`exp04_main_h1`). 240 cells, 0 failed, 23.0 seconds on CPU. This is the source of all headline numbers.

Ablations. `exp05`: quantile slice and importance resampling. `exp06`: calibration-set size (killed primary H2). `exp07`: shift trailing columns.

Table 1: Seed-pooled ΔECE at feature-perturbation $s{=}1.5$. "std" is heterogeneity, not se of the mean. The $n = 60$ rows are 4 datasets $\times$ 3 models $\times$ 5 seeds (pseudo-replicated). Sklearn/synthetic rows are uncalibrated subsets. One-sided Wilcoxon p-values on this pool are exploratory and live in `results/stats_tests.json` (22 tests, unadjusted).

Calibrator	mean ΔECE	std	#pos/neg
none	0.113	0.149	51/9
none (sklearn)	0.042	0.039	25/5
none (synthetic)	0.184	0.183	26/4
temperature	0.124	0.150	56/4
isotonic	0.141	0.151	53/7
histogram	0.141	0.151	57/3

Sanity. I.i.d. logistic regression on breast-cancer: accuracy 0.979, ECE 0.032±0.010(linear models on this set are typically $\gtrsim$ 0.95 accurate). Wine random forests are overconfident i.i.d. (ECE 0.147±0.020), matching the classical tree-calibration picture [2]. Split conformal coverage at $s{=}0$ is near the nominal $1 - \alpha = 0.9$ for $\alpha{=}0.1$. We do *not* compare ECE magnitudes to ImageNet temperature scaling [3].

5 Results

5.1 Feature perturbation usually raises ECE

Figure 1 plots shifted ECE against perturbation strength s. Figure 2 shows mean ΔECE at $s{=}1.5$. Pooled over 60 dataset×model×seed cells, uncalibrated mean ΔECE $=$ 0.113 with cell-to-cell sd 0.149 (51 positive, 9 negative). The same cells have mean ΔBrier 0.257, mean ΔNLL 0.792, and mean accuracy change -0.156. Sklearn-only cells (breast-cancer and wine) have mean ΔECE $= 0.042$ (25/30 positive); synthetics have mean 0.184 (26/30 positive). After i.i.d. temperature scaling the pooled mean ΔECE is 0.124 (sd 0.150). These sd figures describe heterogeneity, not a confidence interval for the mean. Averaging seeds first, all 12 of 12 *uncalibrated* dataset×model means are positive. An exploratory one-sided Wilcoxon on those 12 means gives p=2.4e-04, but models that share a dataset are not independent; a dataset-level one-sided sign test on 4 dataset means is p=0.0625. Breast-cancer histogram boosting is the smallest cell (mean 0.002, seed sd 0.009). Synthetic logistic regression is the largest (mean 0.292, sd 0.214) and coincides with a large accuracy drop. Table 1 records seed-pooled sign counts, not confirmatory p-values; `results/stats_tests.json` contains 22 related one-sided tests with no multiplicity correction.

5.2 Post-hoc maps are not a reliable remedy

Figure 3 plots $\mathrm{ECE}_{\mathrm{shift}}(\mathrm{cal}) - \mathrm{ECE}_{\mathrm{shift}}(\mathrm{none})$. Wine random forests *benefit* from isotonic (-0.123 ECE; shifted ECE 0.175 $\rightarrow$ 0.052), because the uncalibrated forest was already badly calibrated. Temperature scaling on the same cell reaches shifted ECE 0.068. Breast-cancer logistic regression is *hurt* by

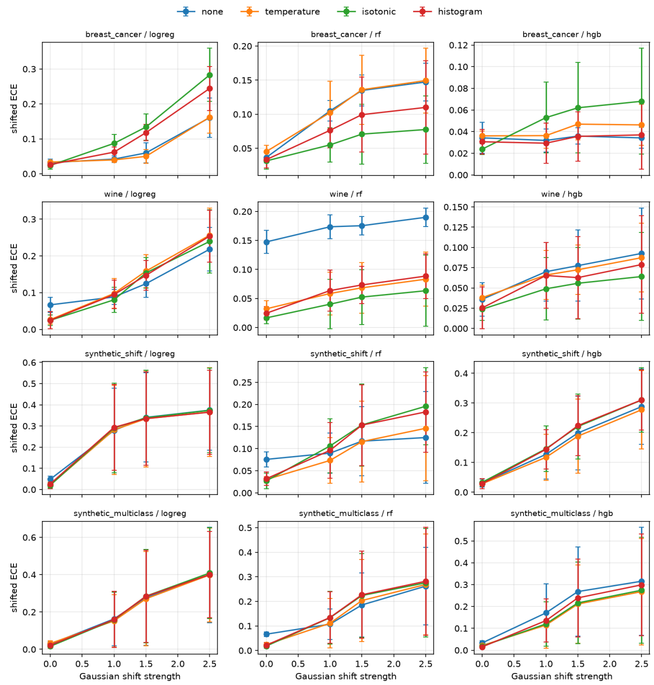

Figure 1: Shifted ECE vs. Gaussian feature-perturbation strength (mean $\pm$ seed sd). s=0 is i.i.d. Labels are frozen; this is not covariate shift. Source: `results/exp04_main_h1.json`.

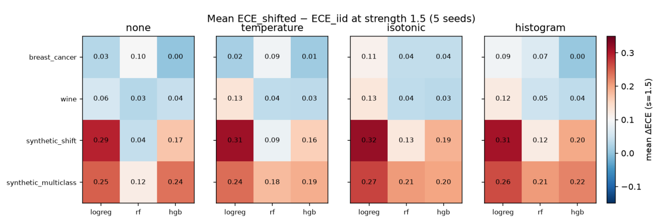

Figure 2: Mean ΔECE at s=1.5. Positive (red) means the shifted test is *worse* calibrated than the i.i.d. test.

isotonic (0.074). We therefore do not rank calibrators under shift.

5.3 Coverage under perturbation

Unweighted split conformal (logreg, `none`, LAC score $1 - p_y$) coverage at $\alpha = 0.1$ (seed mean $\pm$ sd): breast-cancer 0.913$\pm$0.015 (s=0), 0.778$\pm$0.045 (s=1.5), 0.608 (s=2.5); synthetic binary 0.634$\pm$0.190 at s=1.5 (Figure 4). Exchangeability is broken by construction. A 1-D "oracle" density ratio on feature 0 did *not* restore coverage in the H3 pilot (killed).

5.4 Selection shift is a different, weaker effect

Quantile slicing and importance resampling keep original (X, y) pairs, so $P(Y \mid X)$ is preserved and the mechanism is closer to covariate shift. This ablation uses 6 dataset$\times$model cells on 2 datasets (breast-cancer and `synthetic_shift`; wine and the multiclass synthetic are absent). Mean ΔECE (`none`) is 0.016 (sd 0.006) for quantile slice and 0.011 (sd 0.031) for importance resampling. On the *same* 6 cells, Gaussian perturbation at s=1.5 has cell-mean ΔECE 0.105 (sd 0.109). Quantile slice also moves accuracy less (cell-mean Δacc -0.016 vs. matched Gaussian -0.128), so the gap is not identified as "preserving $P(Y \mid X)$" alone. Importance resampling can even *decrease* ECE on breast-cancer. This control **does not support** a claim that i.i.d. post-hoc calibration fails under covariate shift in general. Trailing-column perturbation on `synthetic_shift` yields RF $\Delta\text{ECE} = 0.009$ vs. 0.041 when perturbing the first four columns.

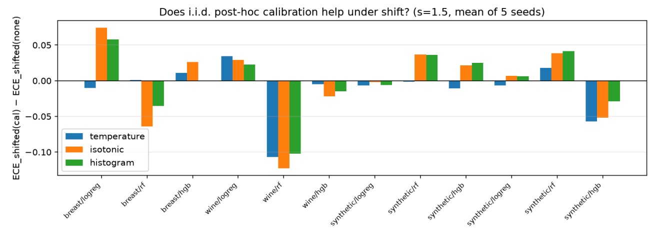

Figure 3: Does i.i.d. calibration help on the shifted test? Negative bars mean the calibrator beats `none` under shift.

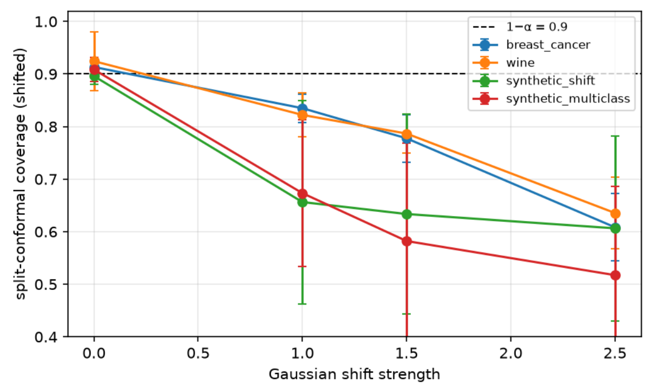

Figure 4: Unweighted split-conformal coverage vs. feature perturbation (logreg, no calibrator, LAC score). Dashed line: $1 - \alpha = 0.9$.

6 Limitations

- **The headline mechanism is not covariate shift.** Frozen-label feature mean-shift changes $P(Y \mid X)$ at the observed x (input corruption). Selection controls that preserve pairings show small ΔECE. Title and abstract are scoped accordingly.
- **Shift is synthetic** and, for feature perturbation, entangled with accuracy drop (synthetic logreg 0.932$\to$0.620 at s=1.5).
- **Small tabular data only.** Wine test splits have 45 points and 44 calibration points with 15 ECE bins; those cells

are noisy. No ImageNet, no LMs, no naturally occurring drift (no Folktables/WILDS).

- **Multiclass isotonic/histogram** map confidence, not a full Dirichlet map [16]; argmax can flip after renormalization.
- **Temperature on probabilities.** Trees have no logits; RF $0/1$ probabilities saturate after clipping. This is not Guo logit temperature.
- **H3 weights were misspecified.** A 1-D histogram ratio is not $p_{\text{test}}(x)/p_{\text{cal}}(x)$ for a 4-D perturbation.
- **Pooling and tests.** Seed-level $n = 60$ Wilcoxon is pseudo-replicated and one-sided. The $n = 12$ seed-averaged uncalibrated means are a sign count, not independent replicates. A dataset-level one-sided sign test has $n = 4$ and p=0.0625. 22 tests, unadjusted. Selection comparisons use 6 cells on 2 datasets and are not strength-matched to s=1.5 perturbation.
- **Numpy grasp proxy, not a robot.** Dual-regime numbers are from `exp08_dual_regime` (HistGradientBoosting, five seeds). No MuJoCo, no DexNet.
- **Search scope.** Novelty is relative to `paper/verified.bib`.

7 Dual-regime act/abort

The tabular measurement says i.i.d. maps fail under frozen-label perturbation, not under pairing-preserving selection. A hiring-relevant task is whether that split changes *what the agent does.*

We use a contact/grasp proxy (`src/dualregime/`): success is physics on true pose, yaw, and force. The *deployed* policy sees only encoder, motor current, and an unused appearance channel. Camera pose and a force gauge are a watchdog. Perturbation biases encoder and motor; labels stay on true physics. Selection restricts true g_x to a far band (pairs kept). A residual detector (encoder$-$camera, motor$-$gauge) labels regimes. Under perturbation the policy *switches channels* and applies a failure-controlling gate $P(\text{act} \mid y{=}0) \approx \alpha$ fitted on the watchdog features. Under i.i.d./selection it keeps the deployed-channel gate. Abort-all and always-watchdog are ablations.

Figure 5 and `results/summary_dual.json`: under encoder/motor bias, router mean utility is 93.0 vs. deployed i.i.d. gating -183.6 vs. abort-all 0.0 (5 seeds). False-confident-act rate 0.044 vs. 0.102; success recall 0.971. On in-support i.i.d. tests the router does not pay a large tax (64.6 vs. 67.6). Far-band selection is hard for every mode; we do not claim the i.i.d. gate transfers out of support. This is a structural proxy, not hardware [22].

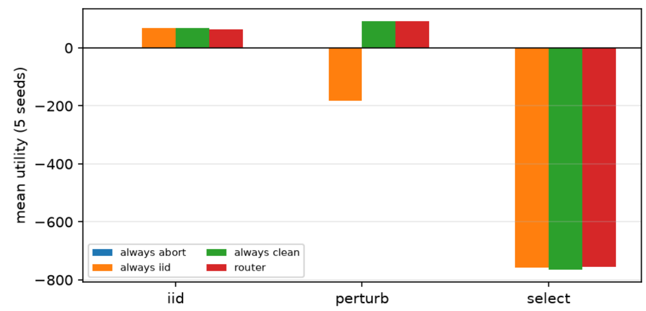

Figure 5: Mean act/abort utility by test split and policy (five seeds). Sensor perturbation is not the same as workspace selection.

8 Conclusion

I.i.d. post-hoc maps are not a dependable answer to frozen-label feature perturbation on these sklearn models, and they are not a dependable *decision* policy when the deployed sensors are the ones that moved. A residual router that switches to watchdog channels recovers utility under that perturbation and is not equivalent to abort-all. Pairing-preserving selection is a different mechanism. The HGB exception and the killed H2/H3 hypotheses remain part of the tabular measurement.

AI-generation disclosure

This repository and draft were produced by a coding agent executing the user's `GOAL.md` protocol (literature verification via arXiv and Semantic Scholar APIs, experiments on CPU, no hand-edited `results/*.json`). Every numeric macro in `paper/numbers.tex` is generated from those files. Citations are restricted to `paper/verified.bib`. A human should still read the limitations and re-run `make fresh-clone-test` before treating the PDF as camera-ready.

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